

Exploring explainable models for point cloud learning

Weiwei Sun

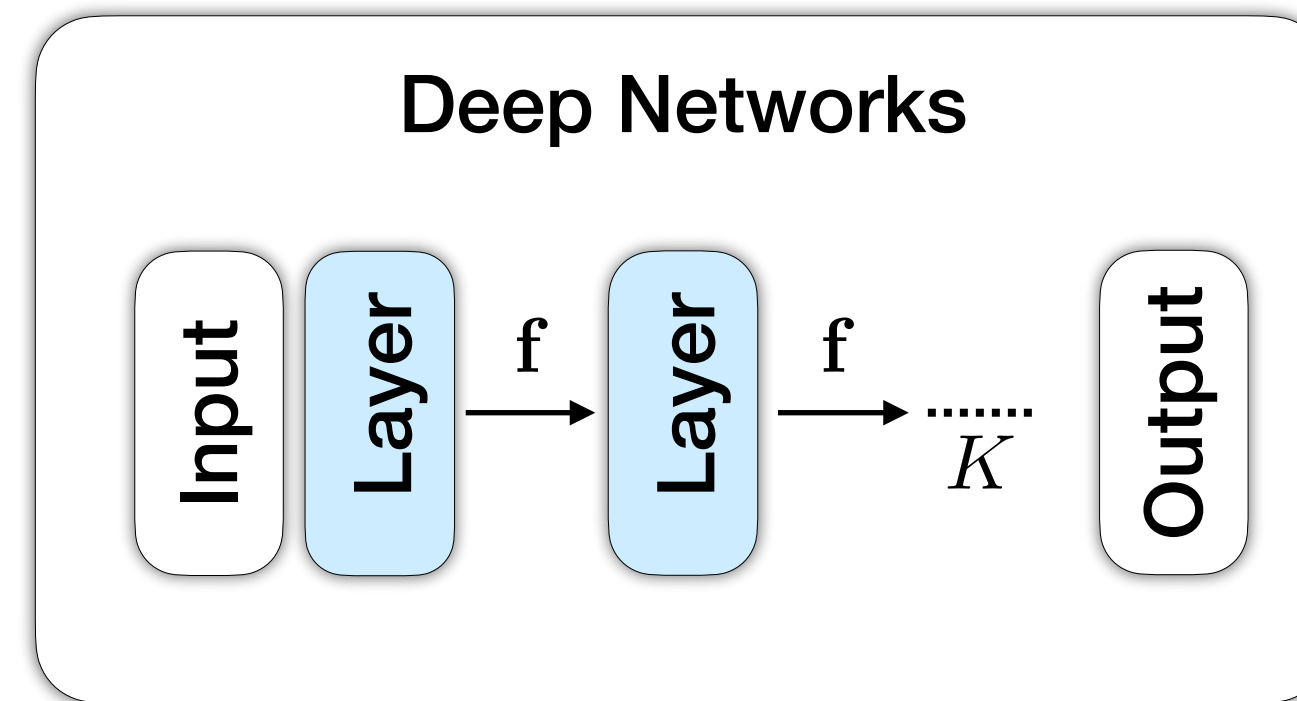
5th Ph.D. student @ UBC

under Dr. Andrea Tagliasacchi & Dr. Kwang Moo Yi



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Explainability is essential



A stack of K layers and a set of intermediate features f .

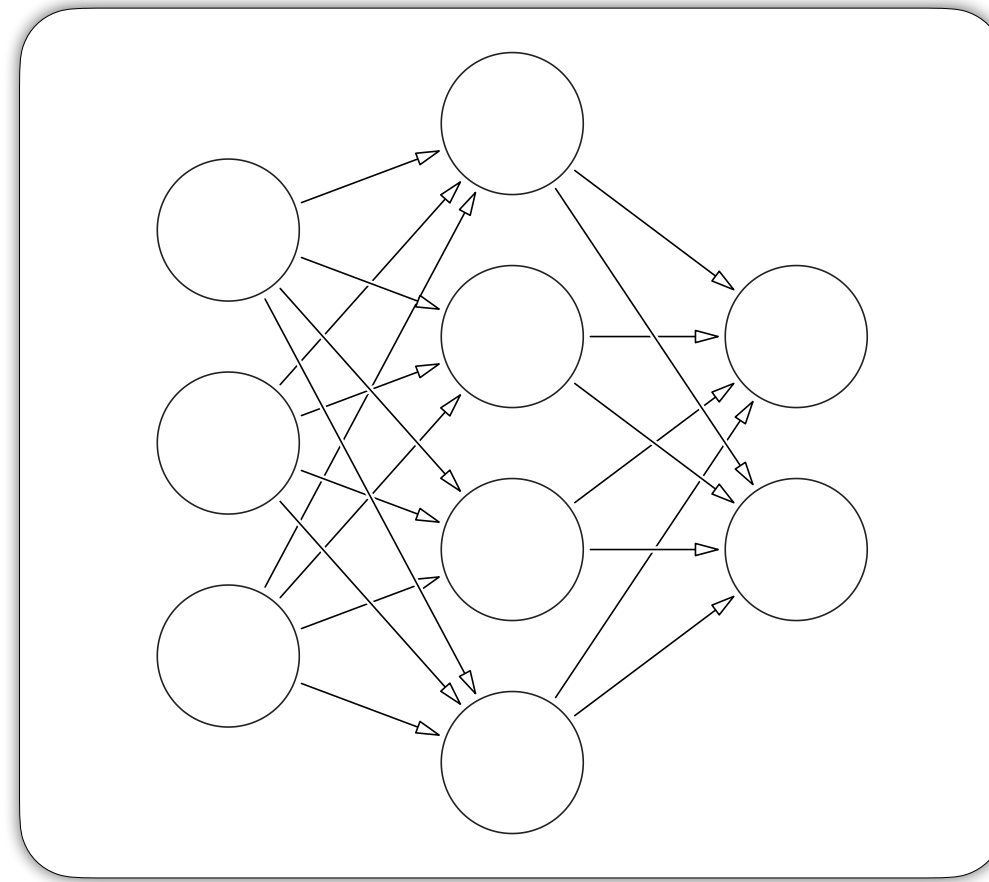
As a black box, deep networks work pretty good
but lacks transparency, controllability, generalization ability.

How does deep networks behave as human understand? And, to what extent, we can benefit from explainable models?

Point cloud learning

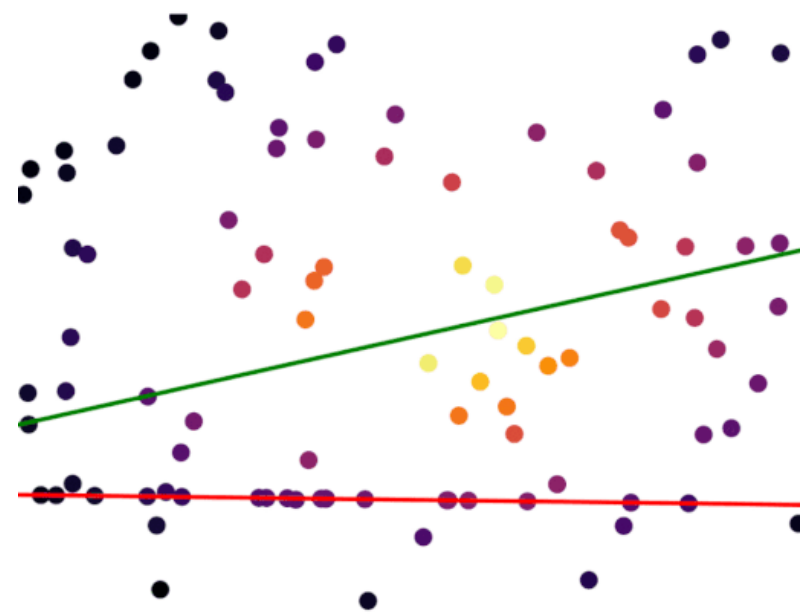
Permutation-equivariant point network

$$\mathbf{P} = \begin{Bmatrix} \mathbf{p}_1 \\ \mathbf{p}_2 \\ \vdots \\ \mathbf{p}_N \end{Bmatrix}$$



$$\mathbf{F} = \begin{Bmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \\ \vdots \\ \mathbf{f}_N \end{Bmatrix}$$

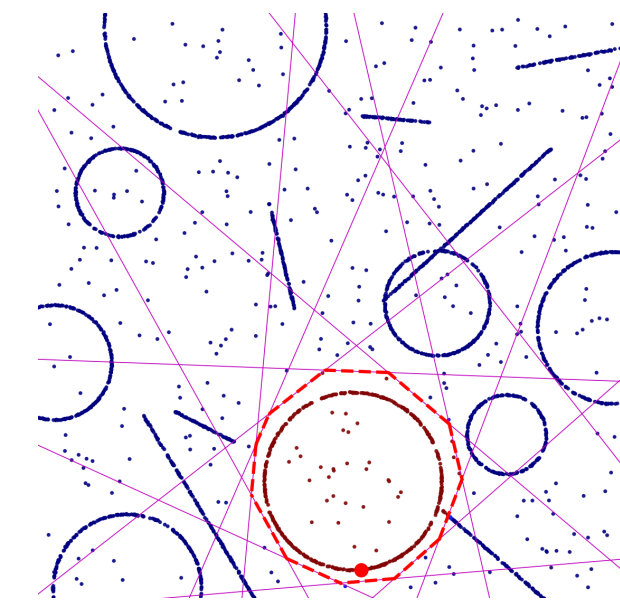
We present models explainable in either layer or features.



ACNe@CVPR2020

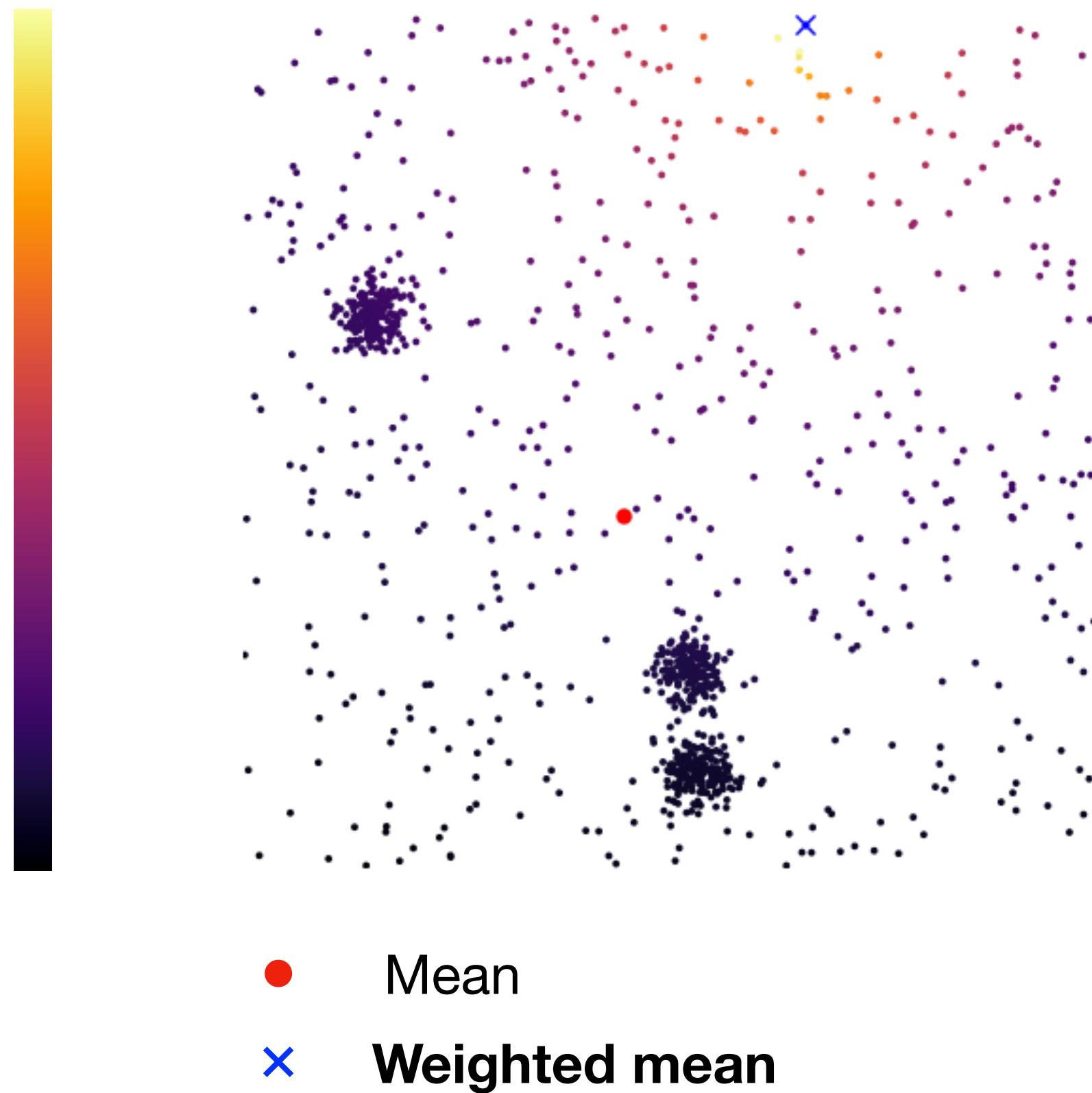


Canonical Capsules@NeurIPS2021

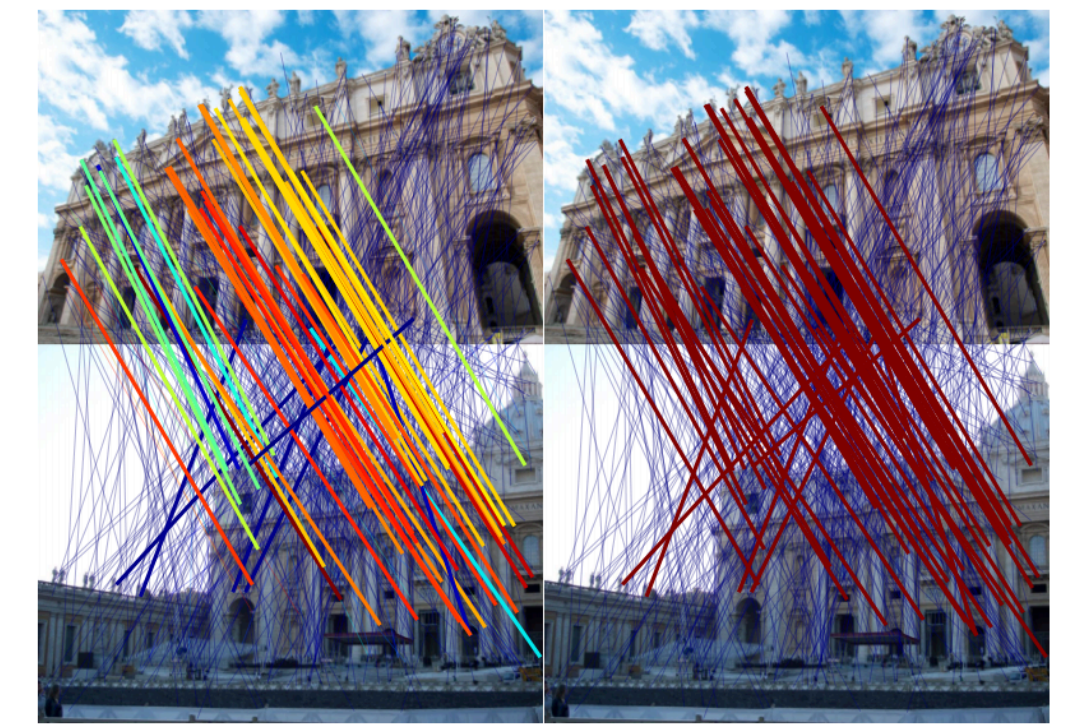
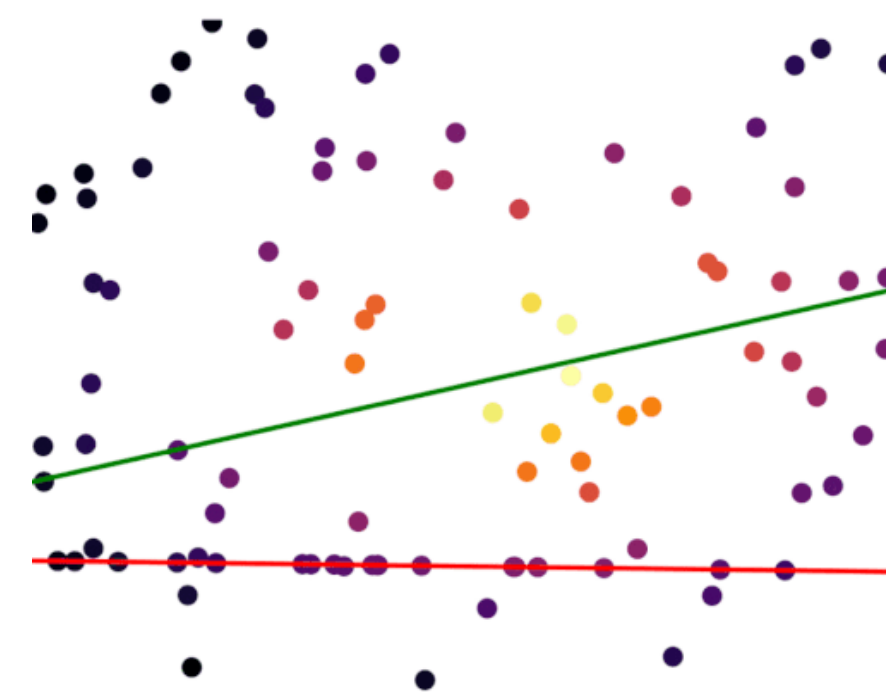
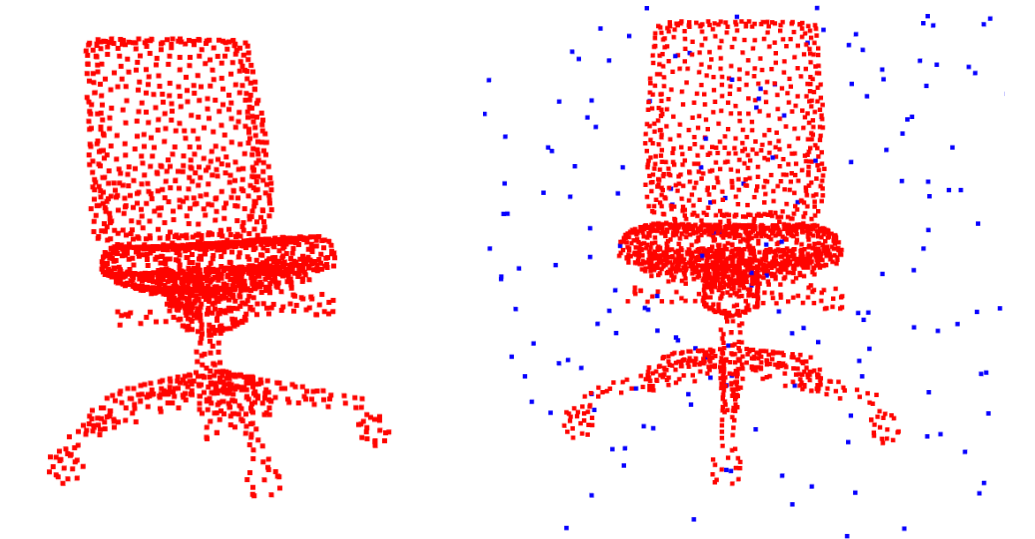
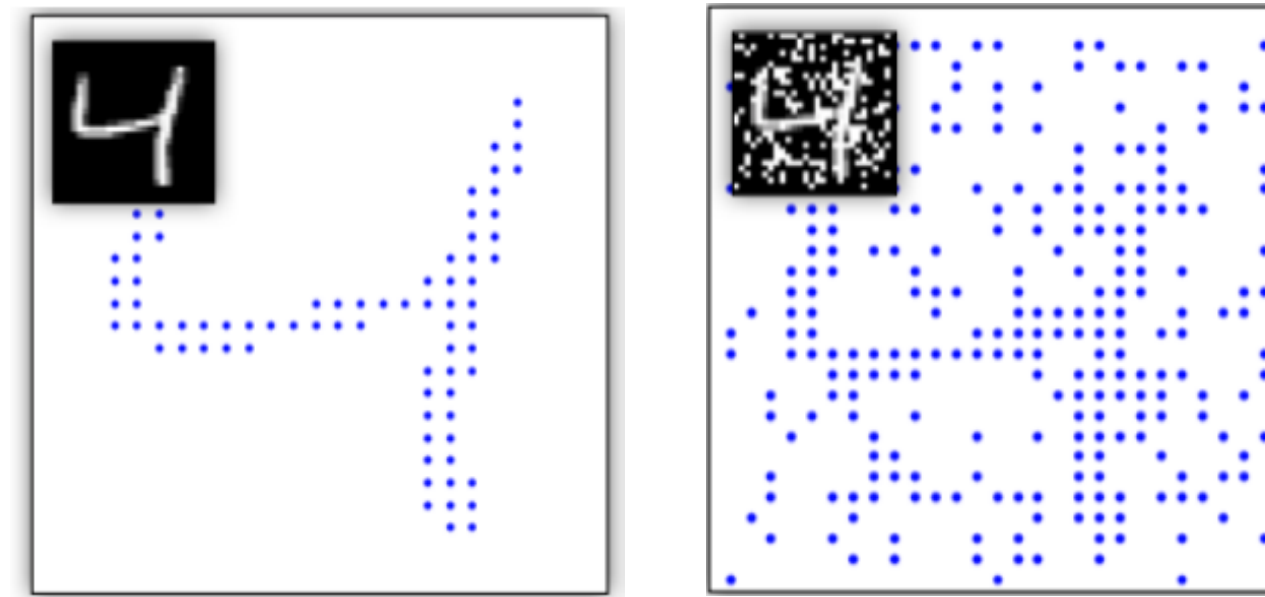


NeuralBF@WACV2023

ACNe: unrolling optimization as deep network



Inspired by IRLS — treating each step as a layer.



Matches: ACN

Matches: Ground Truth

SOTA performance in robust point cloud learning tasks.

Robust Learning of Point Clouds

Feature normalization relates points

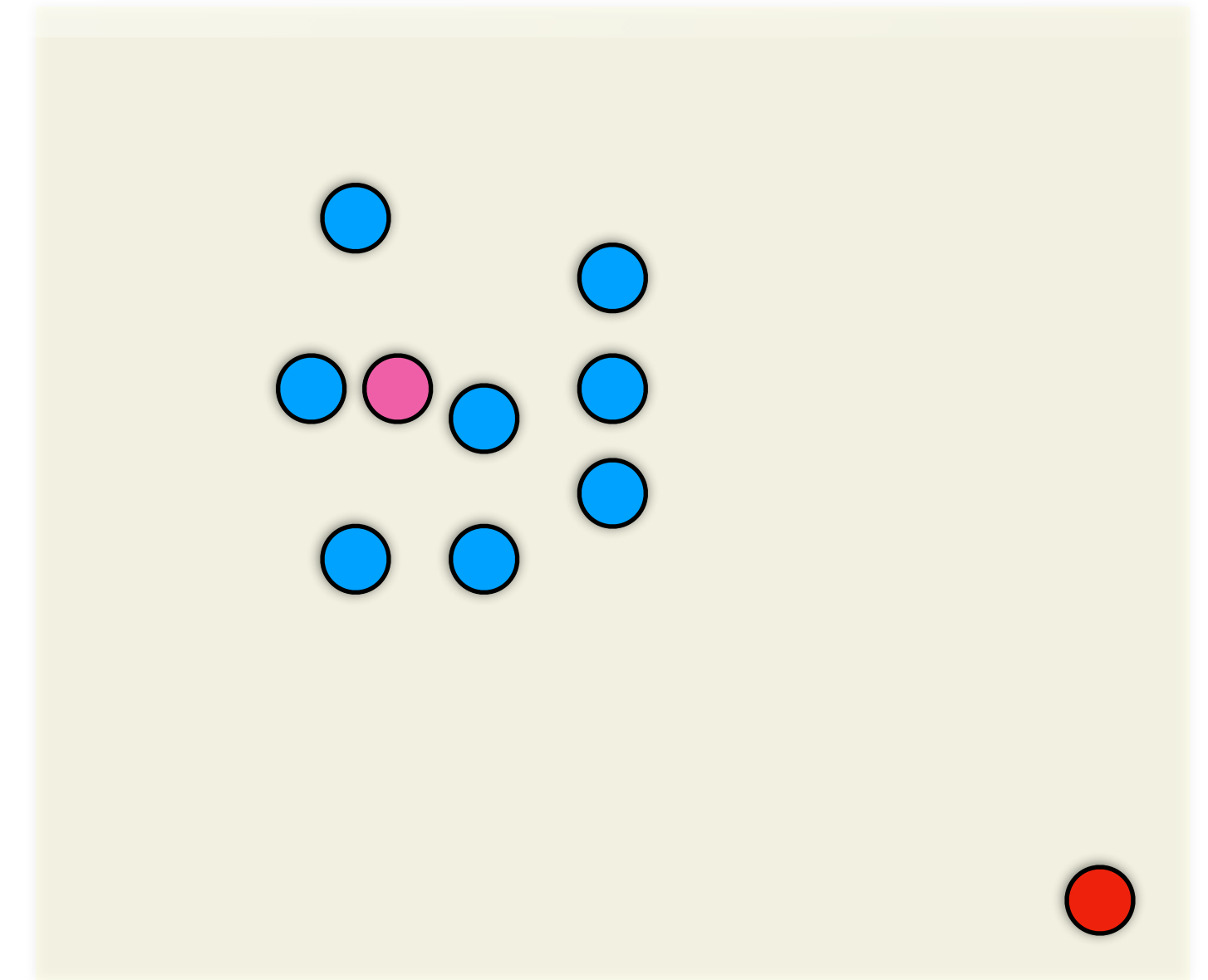
- Context normalization

$$\mathcal{N}_{CN}(\mathbf{f}) = (\mathbf{f} - \mu(\mathbf{f})) \oslash \sigma(\mathbf{f})$$

- Where mean and average can be rewritten as

$$\mu(\mathbf{f}) = \mathbb{E}[\mathbf{f}]$$

$$\sigma(\mathbf{f}) = \sqrt{\mathbb{E}[(\mathbf{f} - \mathbb{E}[\mathbf{f}])^{\circ 2}]}$$



- Mean point
- Input points
- Outlier

Robust Learning of Point Clouds

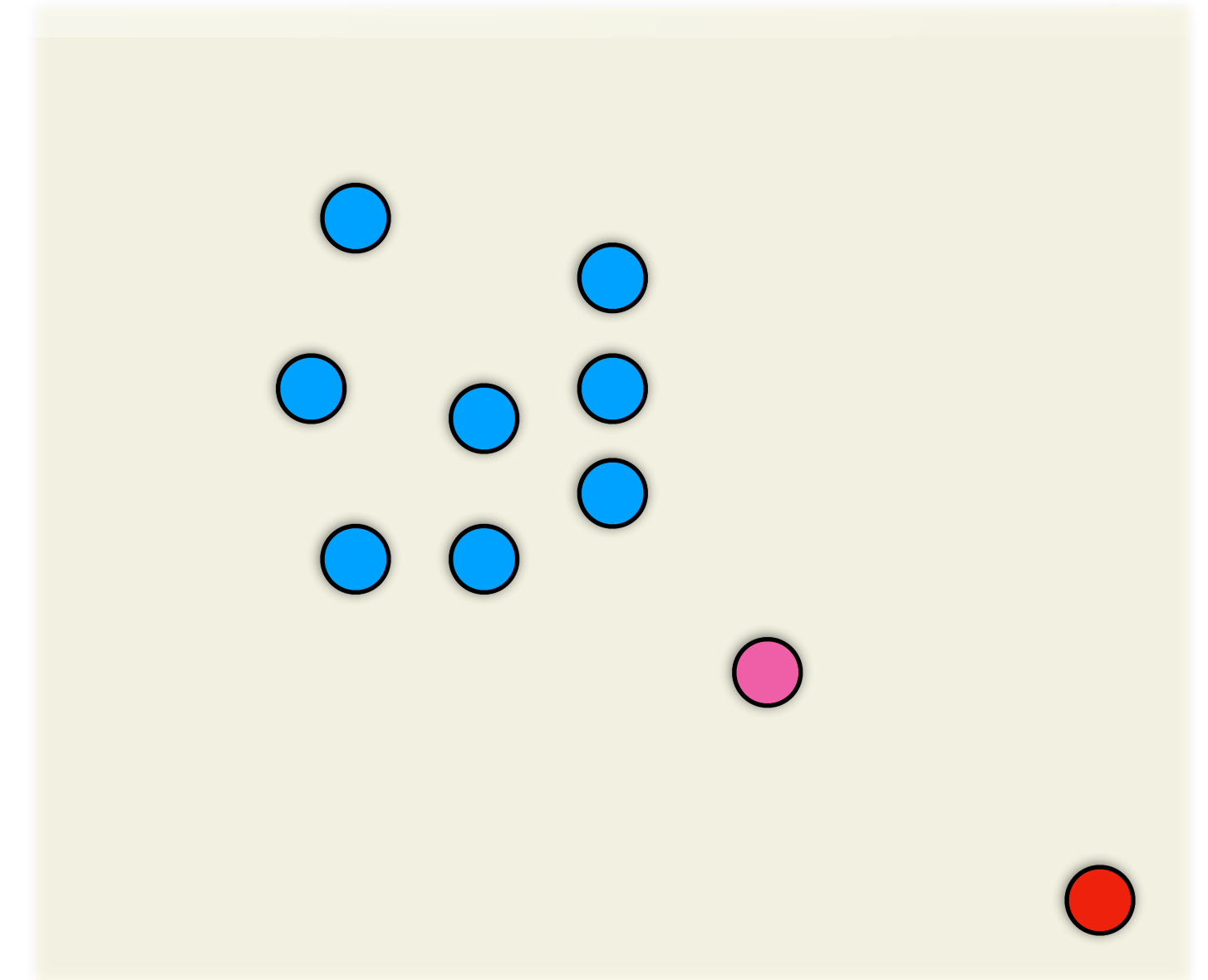
Attentive Context Normalization (ACN)

- So we define our attention based on features

$$\mathbf{w} = \eta(\mathcal{W}_{\omega}(\mathbf{f})), \quad \eta(\mathbf{x}) = \mathbf{x} / \|\mathbf{x}\|_1.$$

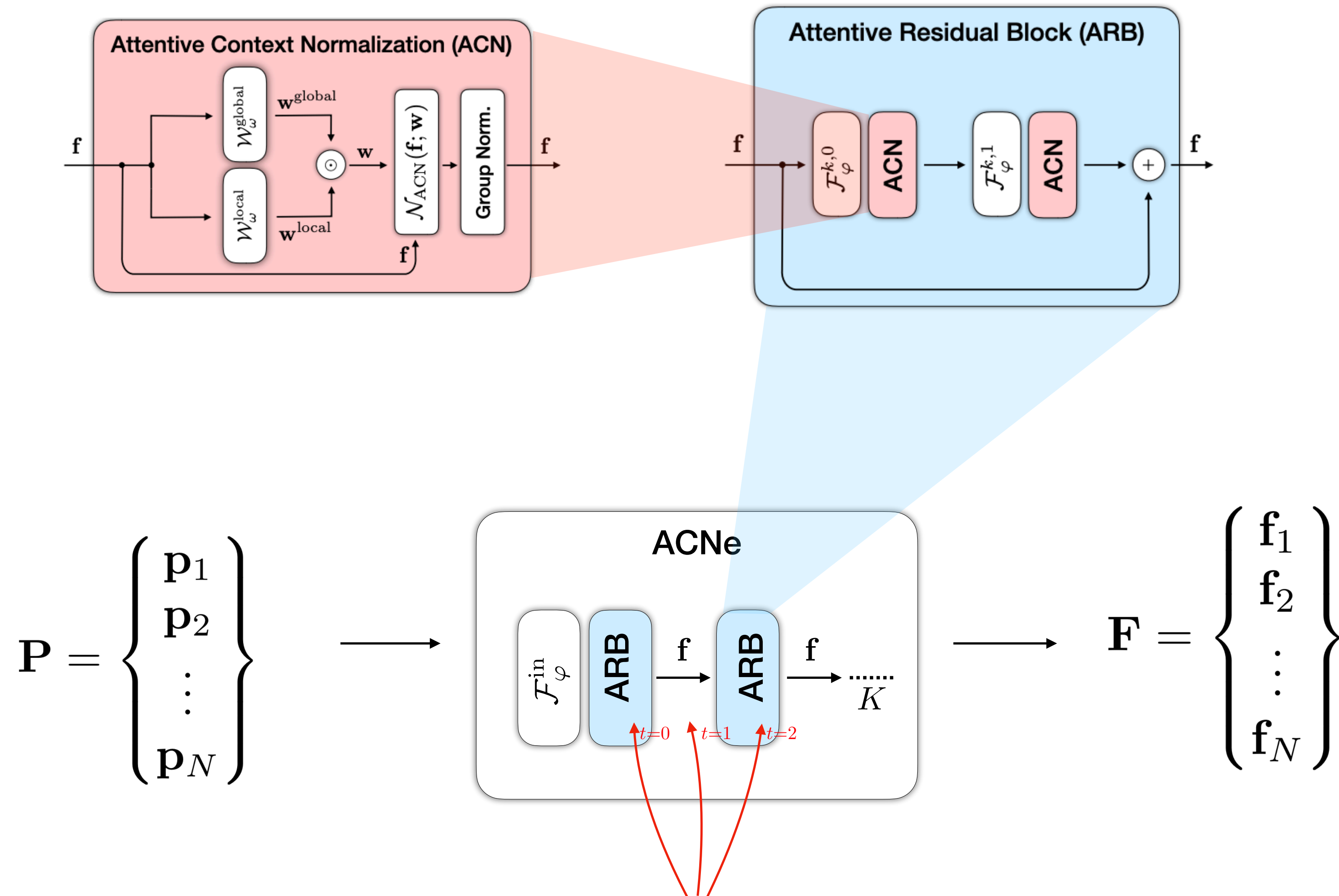
- And then define an attentive normalization

$$\mathcal{N}_{ACN}(\mathbf{f}; \mathbf{w}) = (\mathbf{f} - \mu_{\mathbf{w}}(\mathbf{f})) \oslash \sigma_{\mathbf{w}}(\mathbf{f}).$$



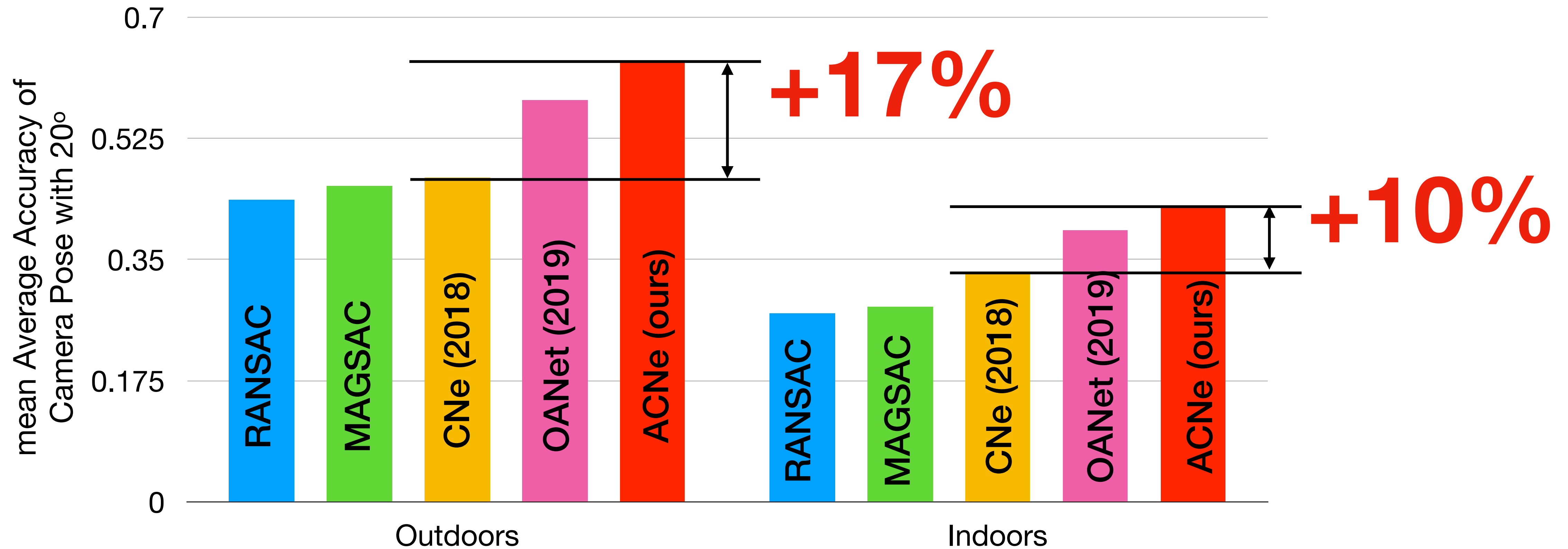
- Mean point
- Input points
- Outlier

ACNe: an iterative network



ACNe: explainable model won with less paramters

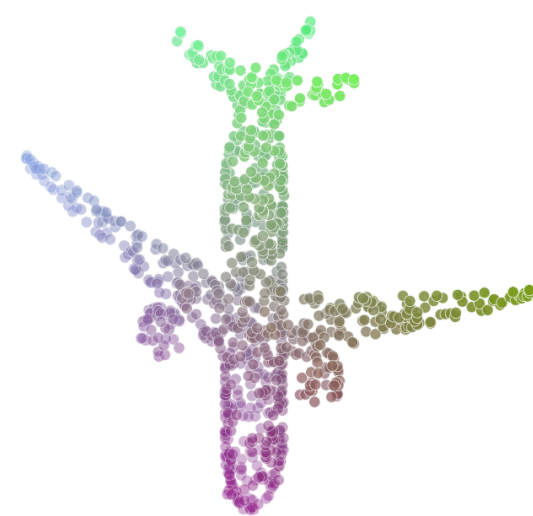
In real-world wide-baseline stereo matching task, explainable ACNe achieves the best performance yet with less parameters.



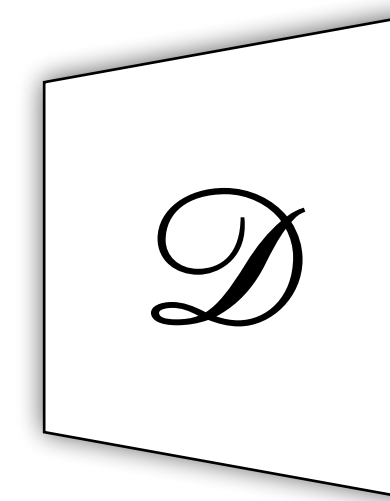
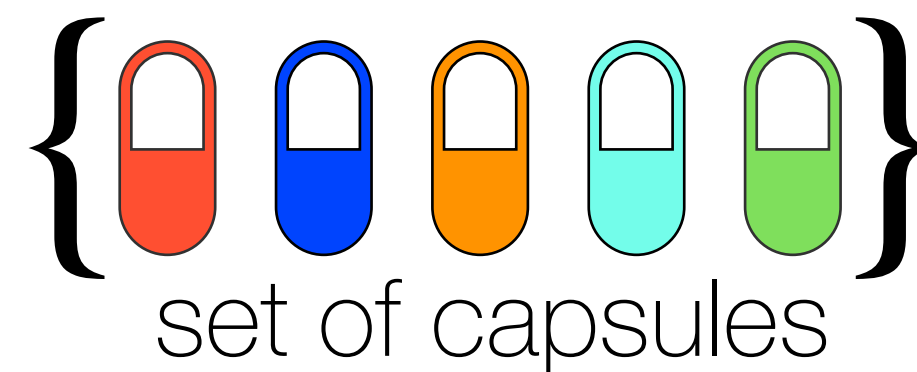
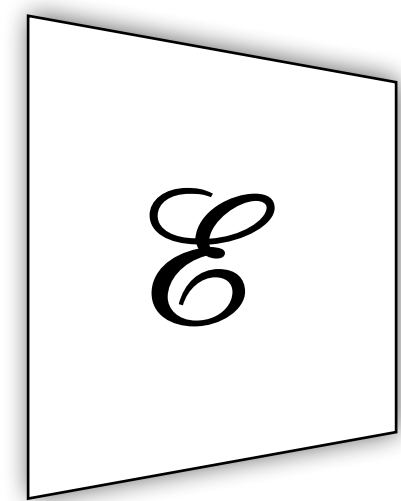
Methods	OANet	CNe	ACNe
# of Parameters	2347K	394K	400K

Canonical Capsules: explainable features

- A **self-supervised** training architecture for **point clouds** with:
 - SOTA performance in **auto-encoding**
 - SOTA performance in unsupervised **classification**
 - SOTA performance in learnt **canonicalization**



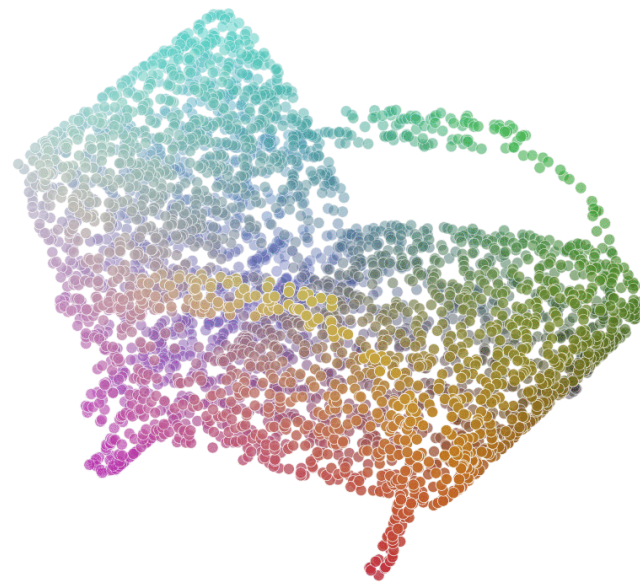
input point cloud
(randomly transformed)



auto-encoding
(learnt canonical frame)

Built-in biases in 3D datasets

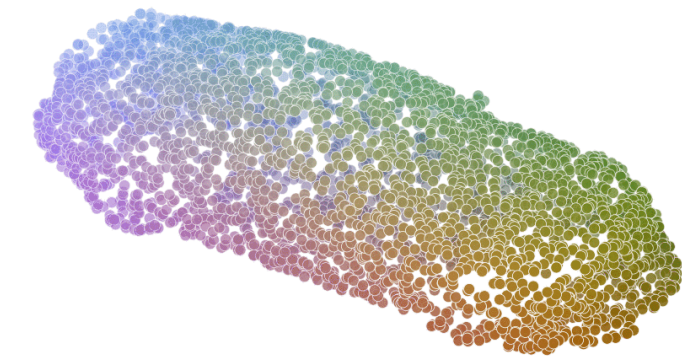
chairs



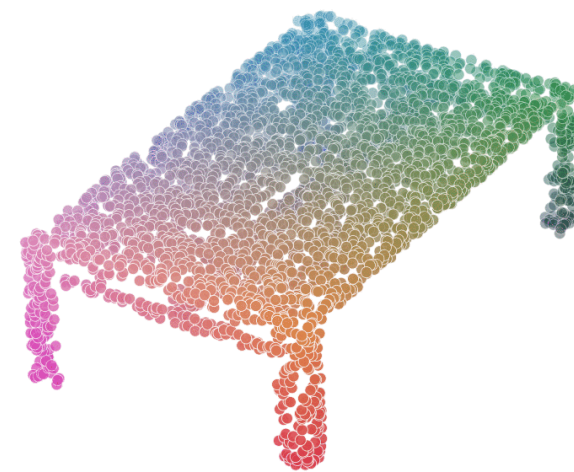
airplane



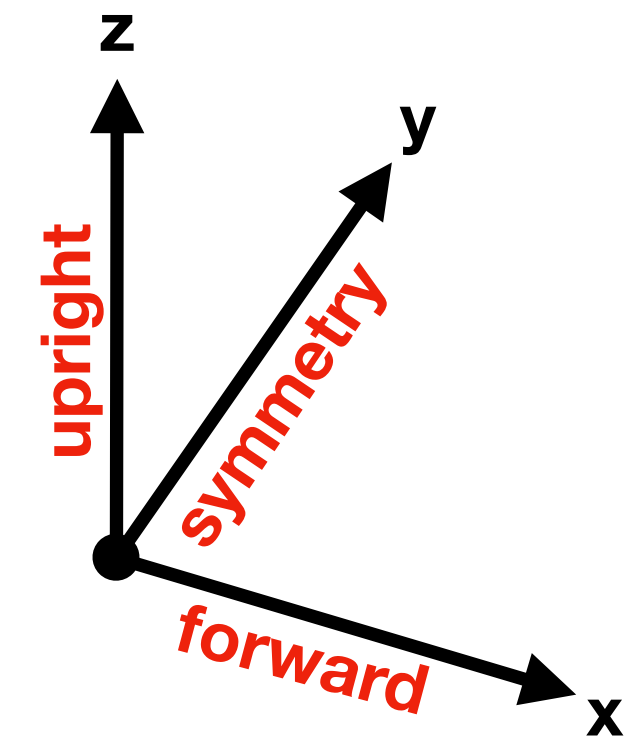
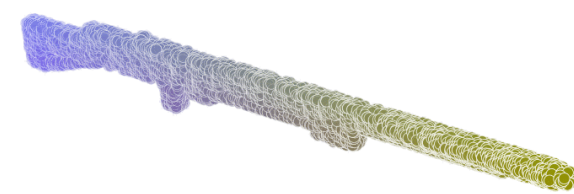
car



table



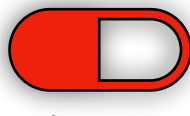
firearm



Auto-encoding Point Clouds

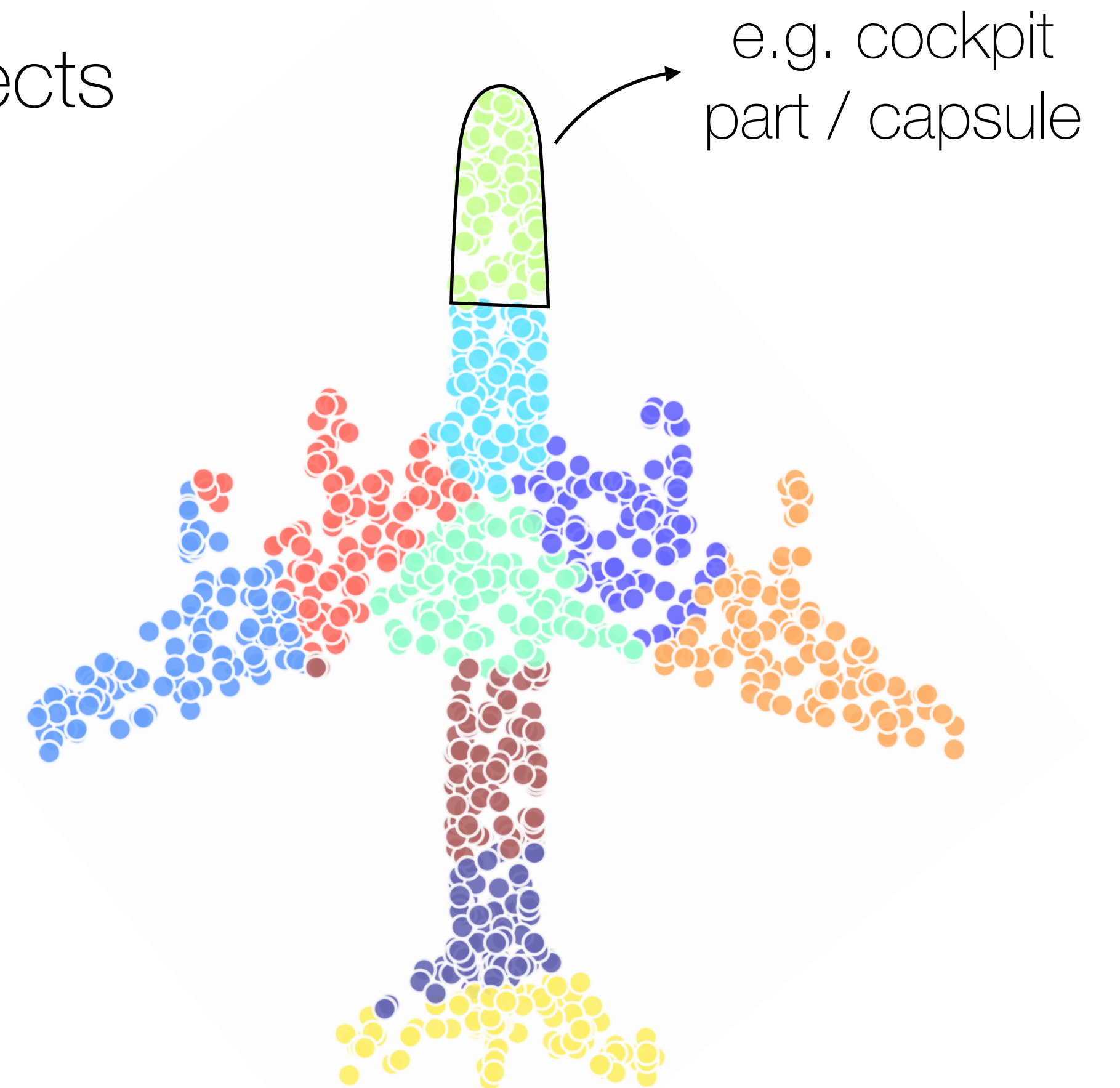


But... what are capsules?

- An abstraction of **scene graphs** from computer graphics
- Capsules are **part-based** representations of objects
- Think of a capsule  as an “oriented part”

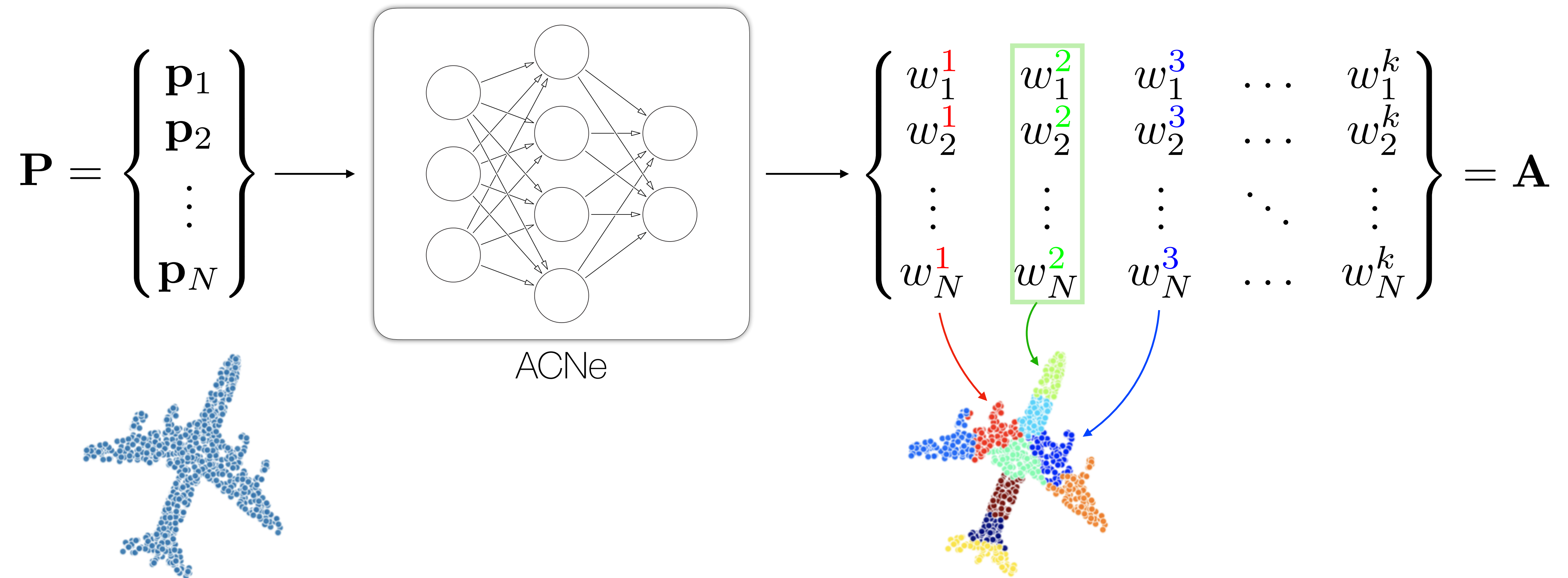
capsule descriptor
part **shape**

capsule pose
part **orientation**



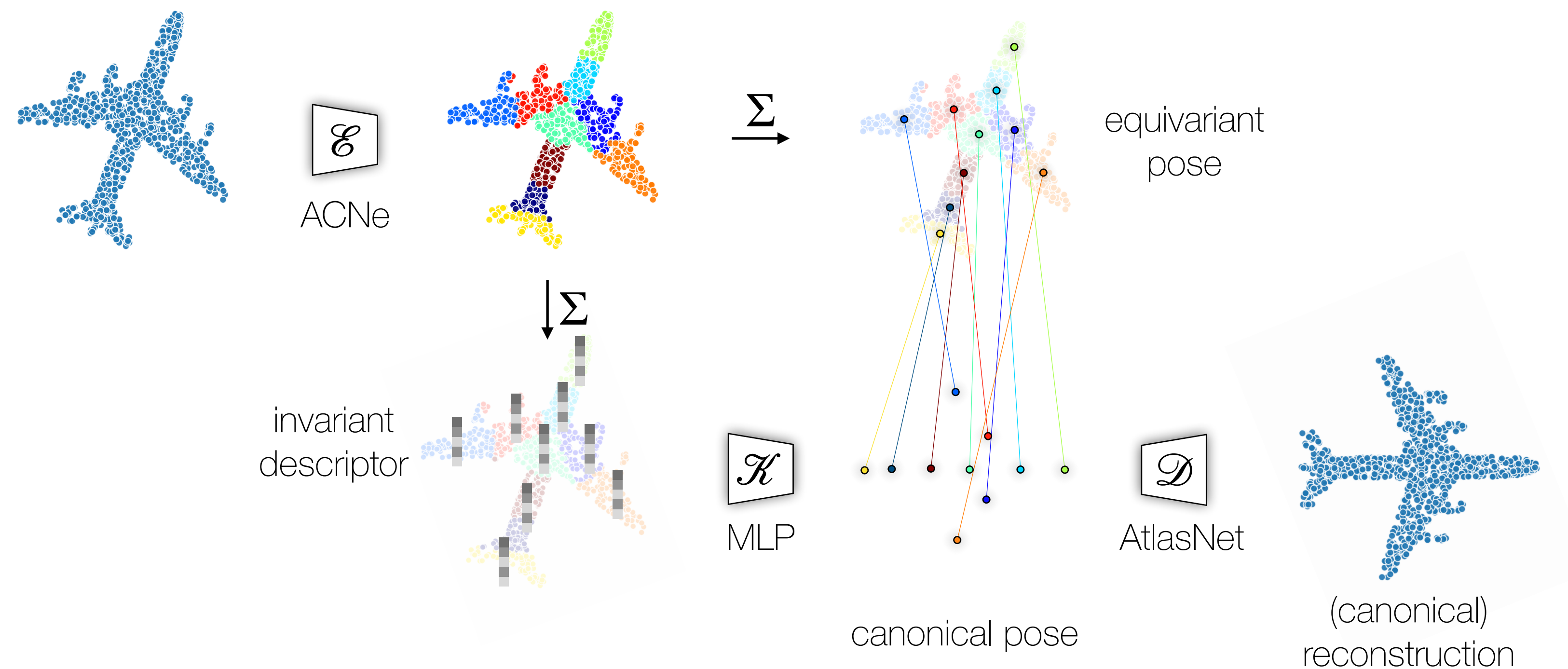
Credit to: Hinton et al. NeurIPS 2011 «Transforming Auto-encoders»

Multi-head attention (ACNe)

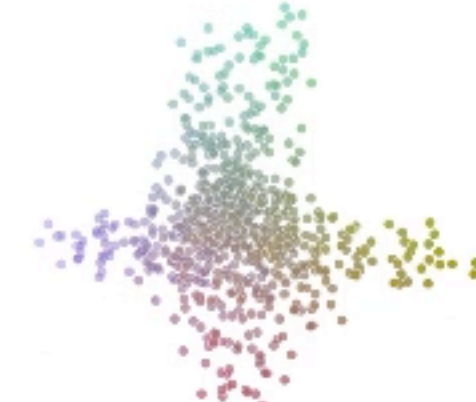
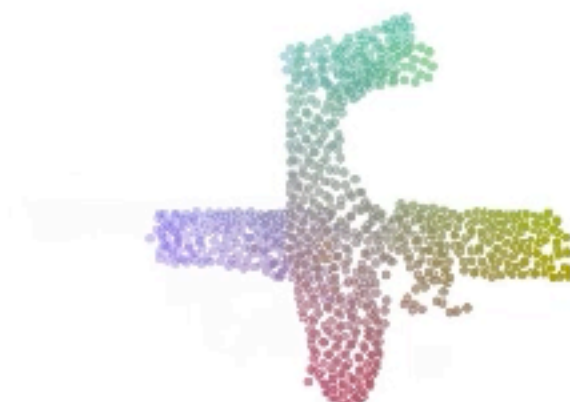
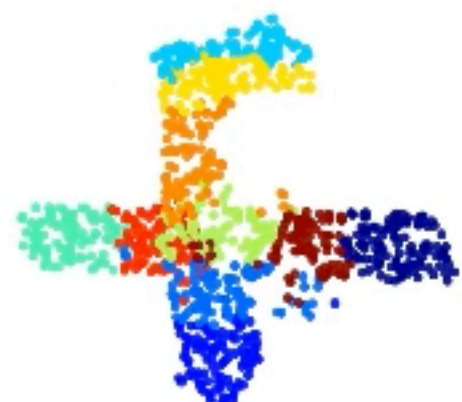
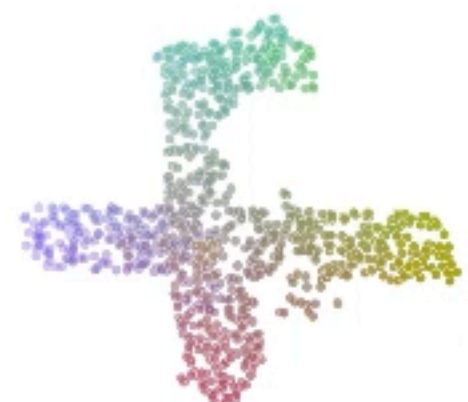
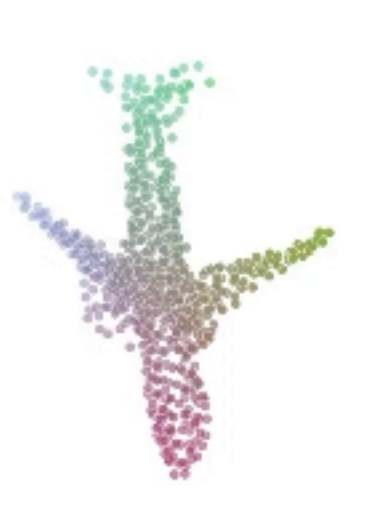
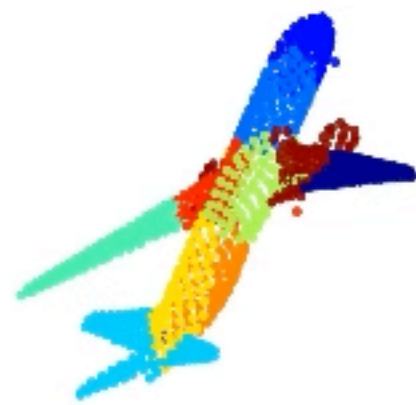
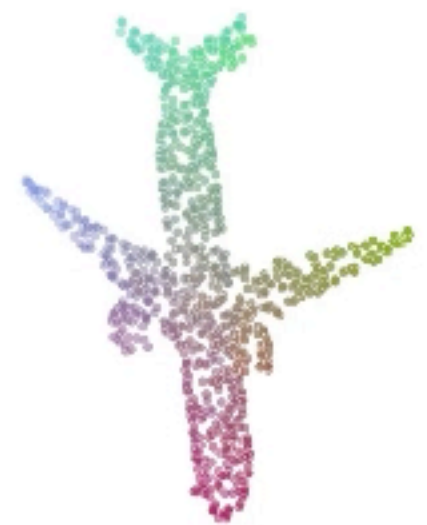
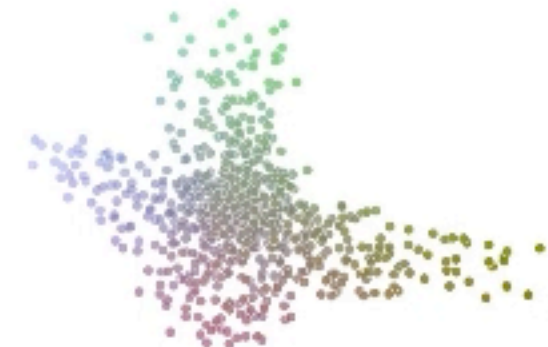
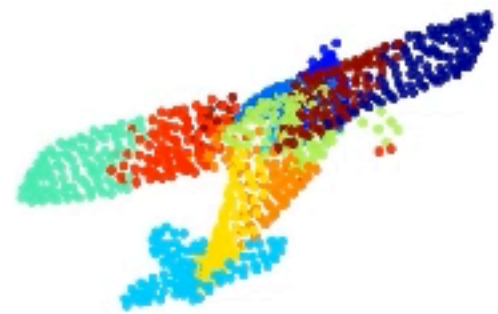
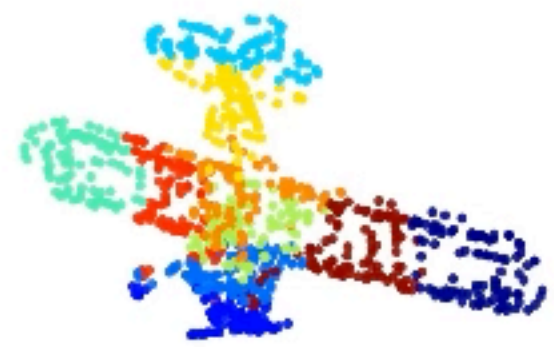
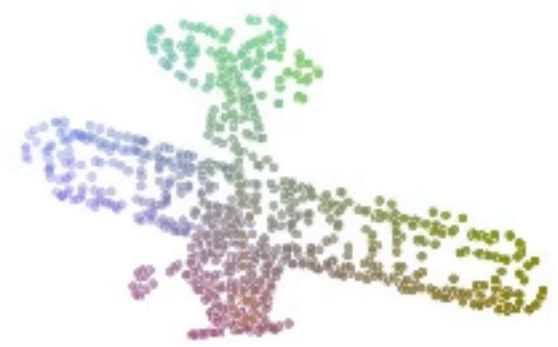


Inference Procedure

- Let us look at how the **inference** (forward pass) of our network architecture
 - Note the **input** is a point cloud in **any pose** (random rotation/translation)
 - While the **reconstruction** is executed in the **canonical frame**



Auto Encoding – Controllability and better performance



Input
(Point Cloud)

Decomposition
(original frame)

Reconstruction
(canonical frame)

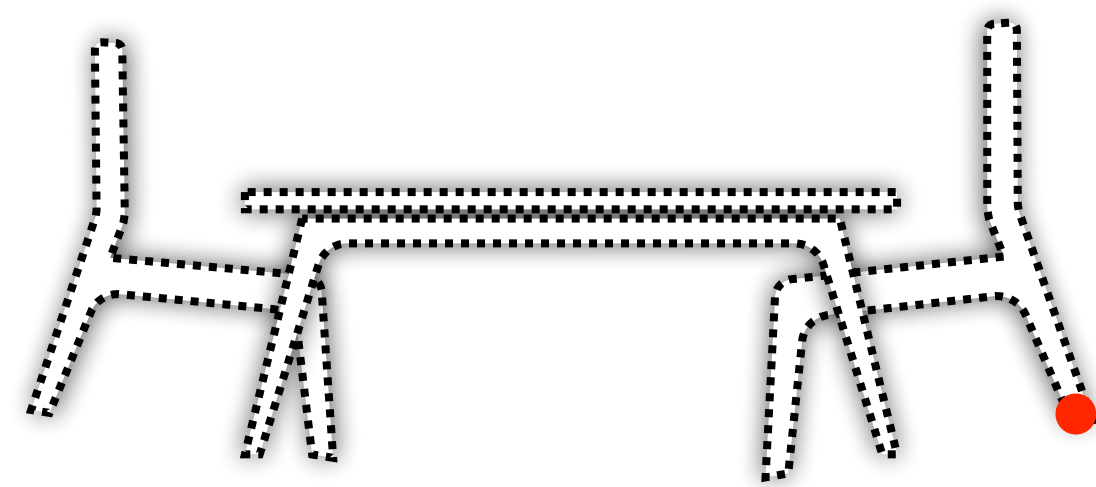
Reconstruction
(original frame)

PointCapsNet
(CVPR 2019)

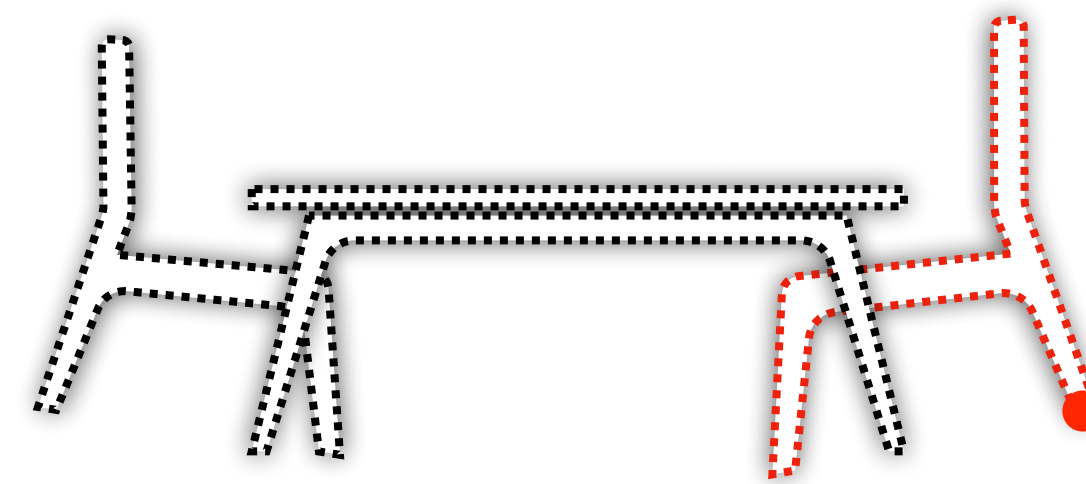
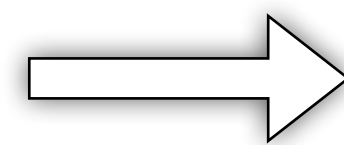
AtlasNetV2
(NeurIPS 2019)

NeuralBF: Explainable feature and layer

- An **iterative** layer that
 - is inspired by traditional **bilateral filtering**
 - generates **query-conditioned** instance proposals for 3D point clouds
 - gives state-of-the-art **top-down** 3D instance segmentation.



Query



Affinity

Direct bounding box regression

- A **commonly used** method for instance proposal generation
- Widely adopted for **top-down** 2D instance segmentation
- However, **bottleneck** the performance of top-down 3D instance segmentation

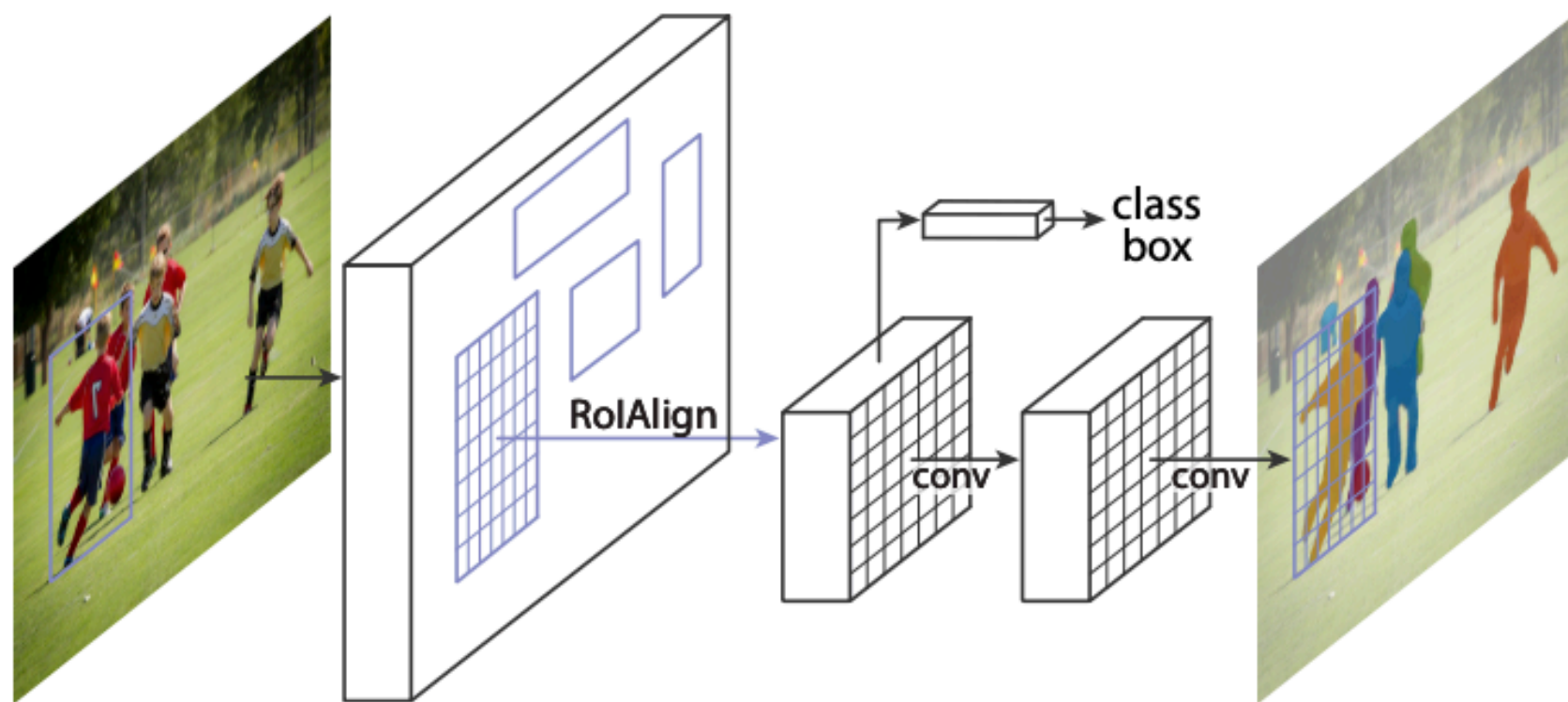
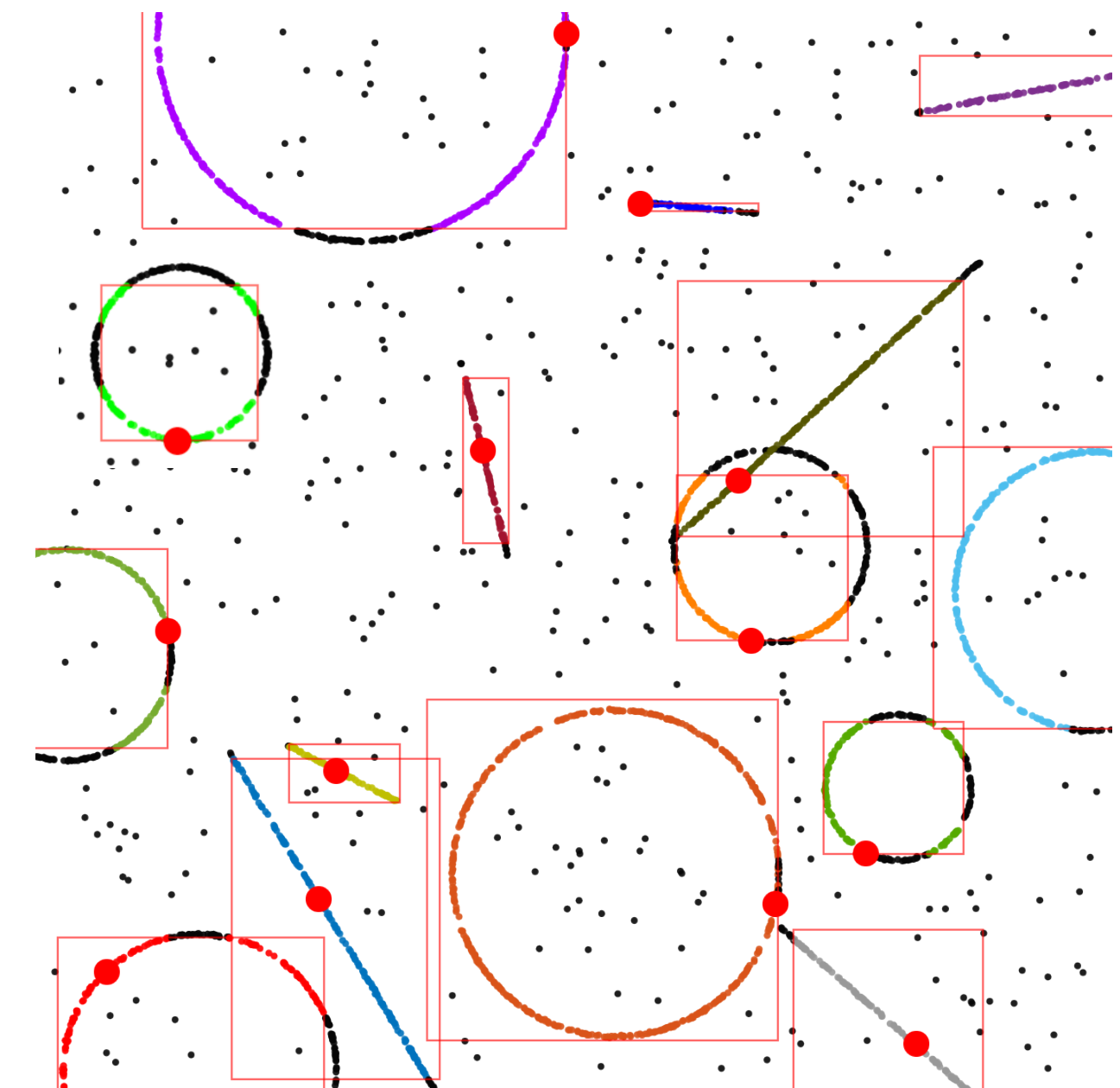
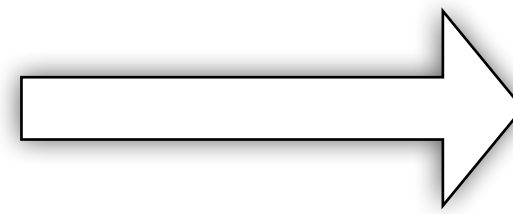


Figure 1. The **Mask R-CNN** framework for instance segmentation.

He et al. Mask R-CNN. ICCV 17.

Great success in 2D images

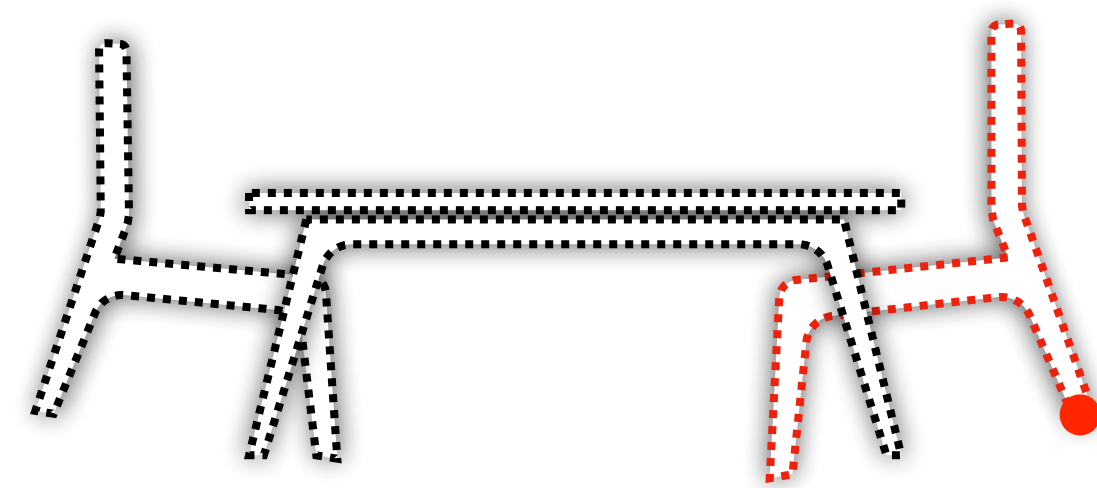
To 3D point cloud



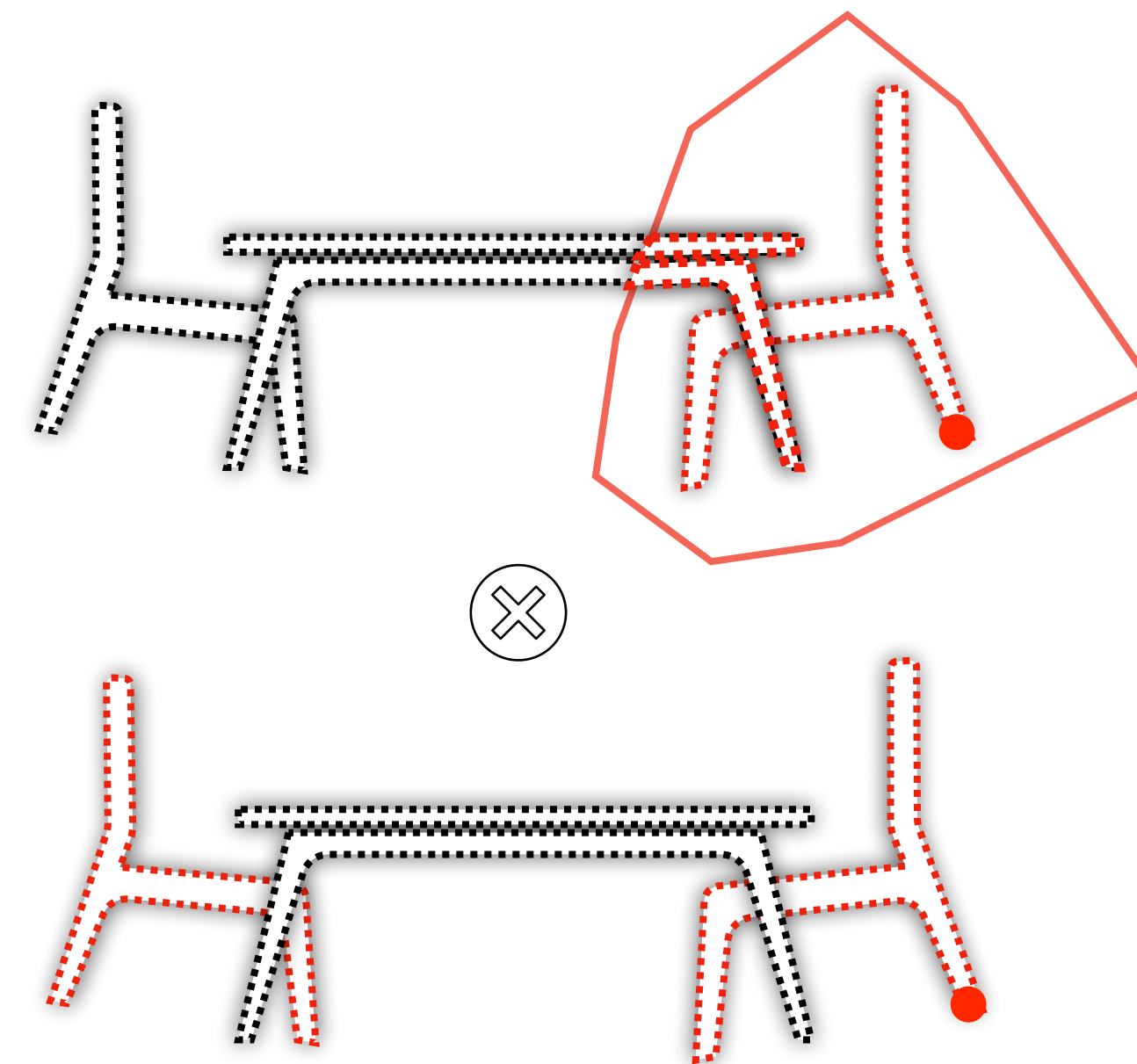
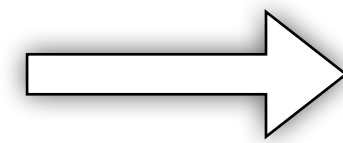
mAP < 50%

Affinity as proposal

- Input *a given query (q)* on the input point cloud.
- Output affinity score map of range $[0, 1]$.
- Consists of *spatial* and *semantic* affinity.



Affinity
 $\mathcal{A}(q)$



$\mathcal{A}_p(q)$
Spatial similarity

Semantic similarity
 $\mathcal{A}_f(q)$

Semantic similarity

- A simple exponential kernel
 - L2 distance between semantic similarity features

$$\mathcal{A}_{\mathbf{f}}(\boxed{q})[\boxed{n}] = \exp(-\boxed{\tau_{\mathbf{f}}} \cdot \boxed{\mathcal{K}_{\mathbf{f}}(q, n)})$$

The diagram illustrates the components of the semantic similarity equation $\mathcal{A}_{\mathbf{f}}(\boxed{q})[\boxed{n}] = \exp(-\boxed{\tau_{\mathbf{f}}} \cdot \boxed{\mathcal{K}_{\mathbf{f}}(q, n)})$. Annotations with arrows point to specific parts of the equation:

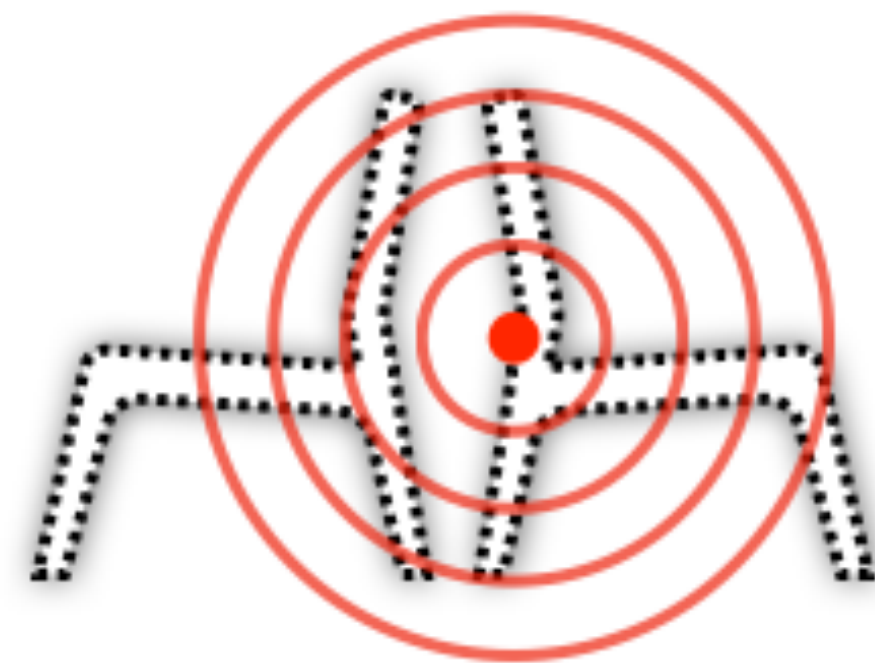
- Query**: Points to the variable q inside the first box.
- Indexing n-th element**: Points to the variable n inside the second box.
- Temperature**: Points to the variable $\tau_{\mathbf{f}}$.
- Backbone feature**: Points to the variable $\mathbf{f}$ inside the kernel function box.
- L2 distance kernel**: Points to the kernel function $\mathcal{K}_{\mathbf{f}}(q, n)$.

Spatial similarity

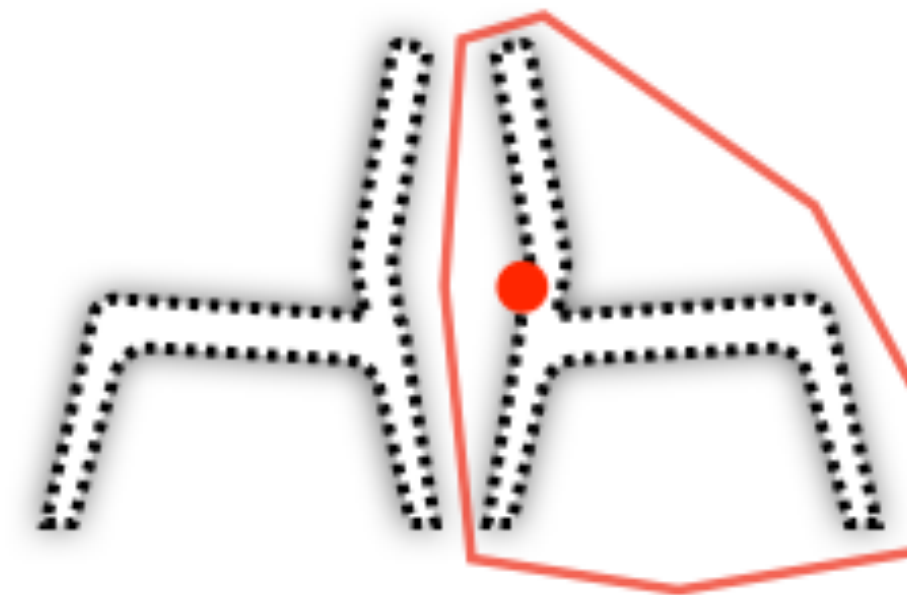
- A non-isotropic kernel based on local convex hulls
 - Is able to isolate the queried instance

$$\mathcal{A}_{\mathbf{p}}(q)[n] = \exp(-\tau_{\mathbf{p}} \cdot \mathcal{K}_{\mathbf{p}}(q, n))$$

Temperature



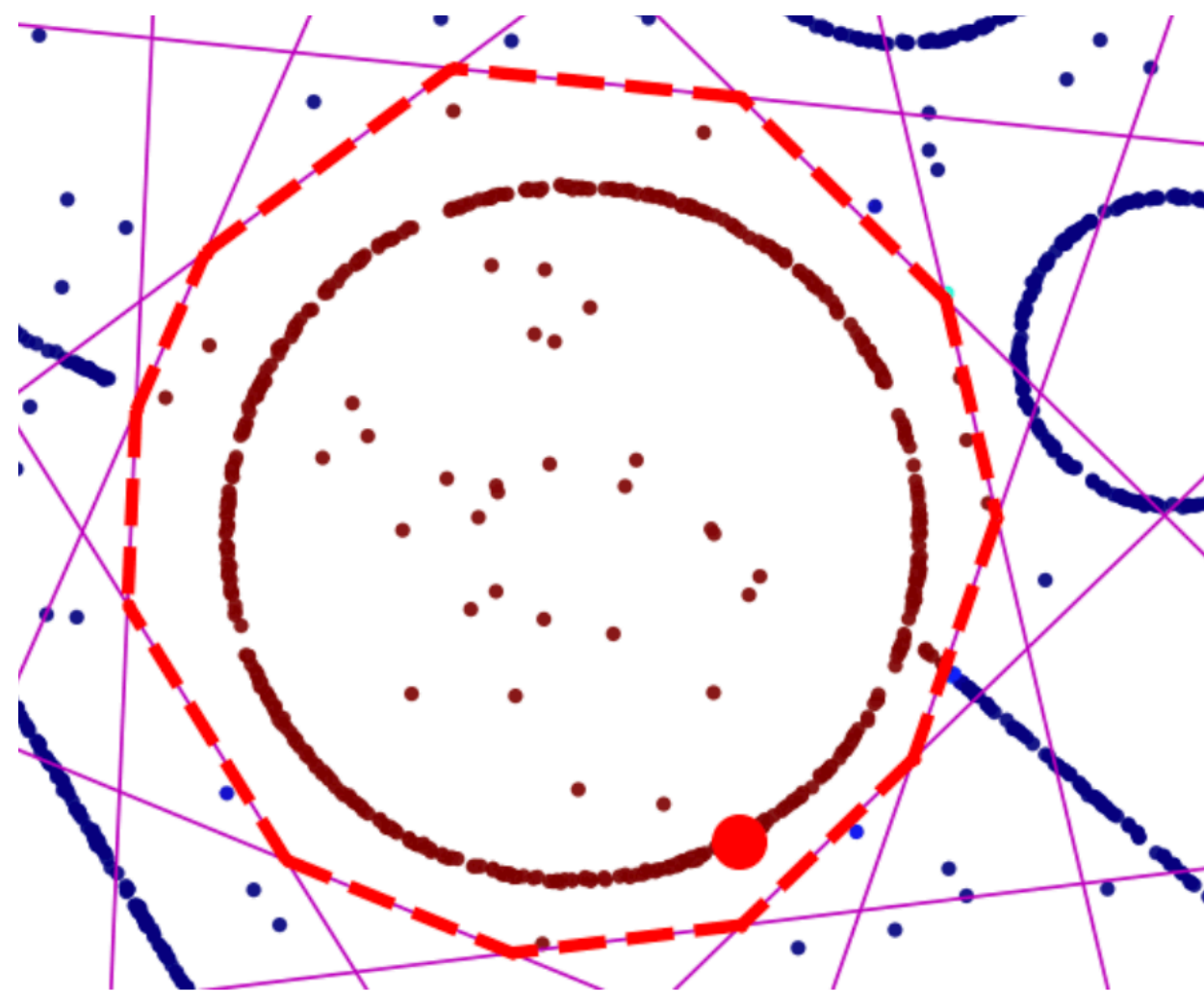
isotropic affinity



non-isotropic affinity

Spatial similarity — in more details

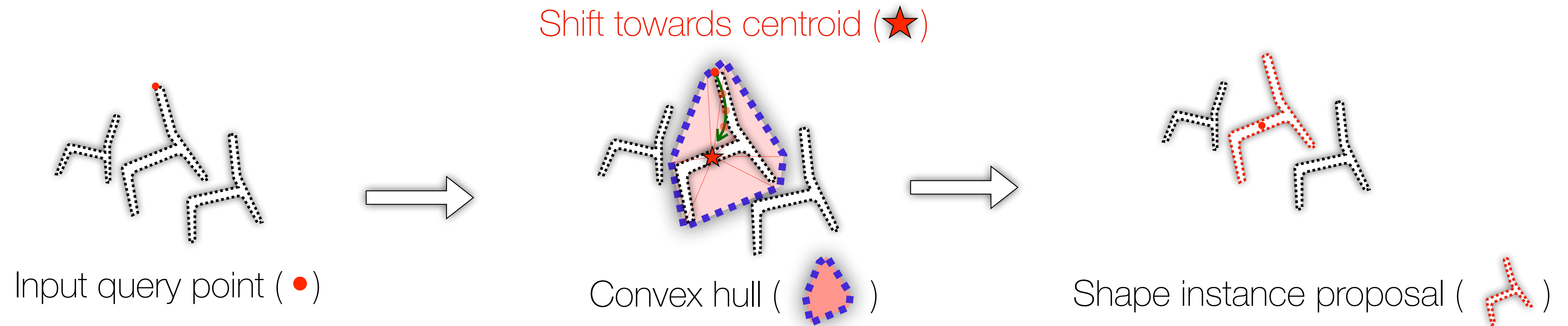
- A **hyper coordinate network** -- convex hull
 - Consists of a set of hyperplanes conditional on query
 - Is an explainable neural field—comparing with recent MLP-based neural field.
 - Is very **small and memory efficient**



$$\mathcal{C}(\mathbf{x}; \mathbf{f}) \begin{cases} = 0 & \text{if } \mathbf{x} \text{ inside convex defined by } \mathbf{f}, \\ > 0 & \text{otherwise } (\approx \text{boundary distance}). \end{cases}$$

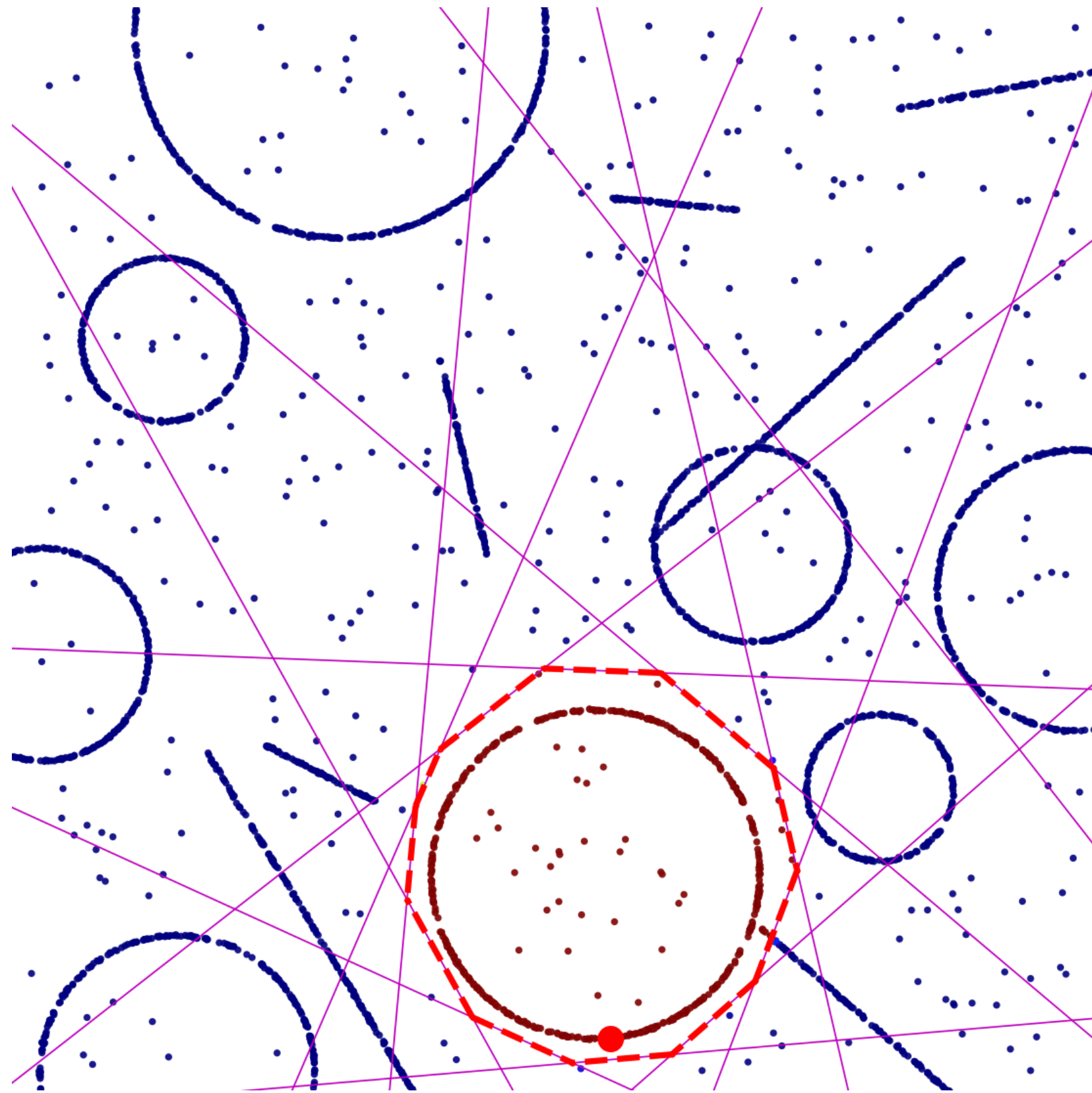
Bilateral filtering

- Iterative inference
 - Update query as the **weighted center** of input
 - Shift query into the centroid of instance

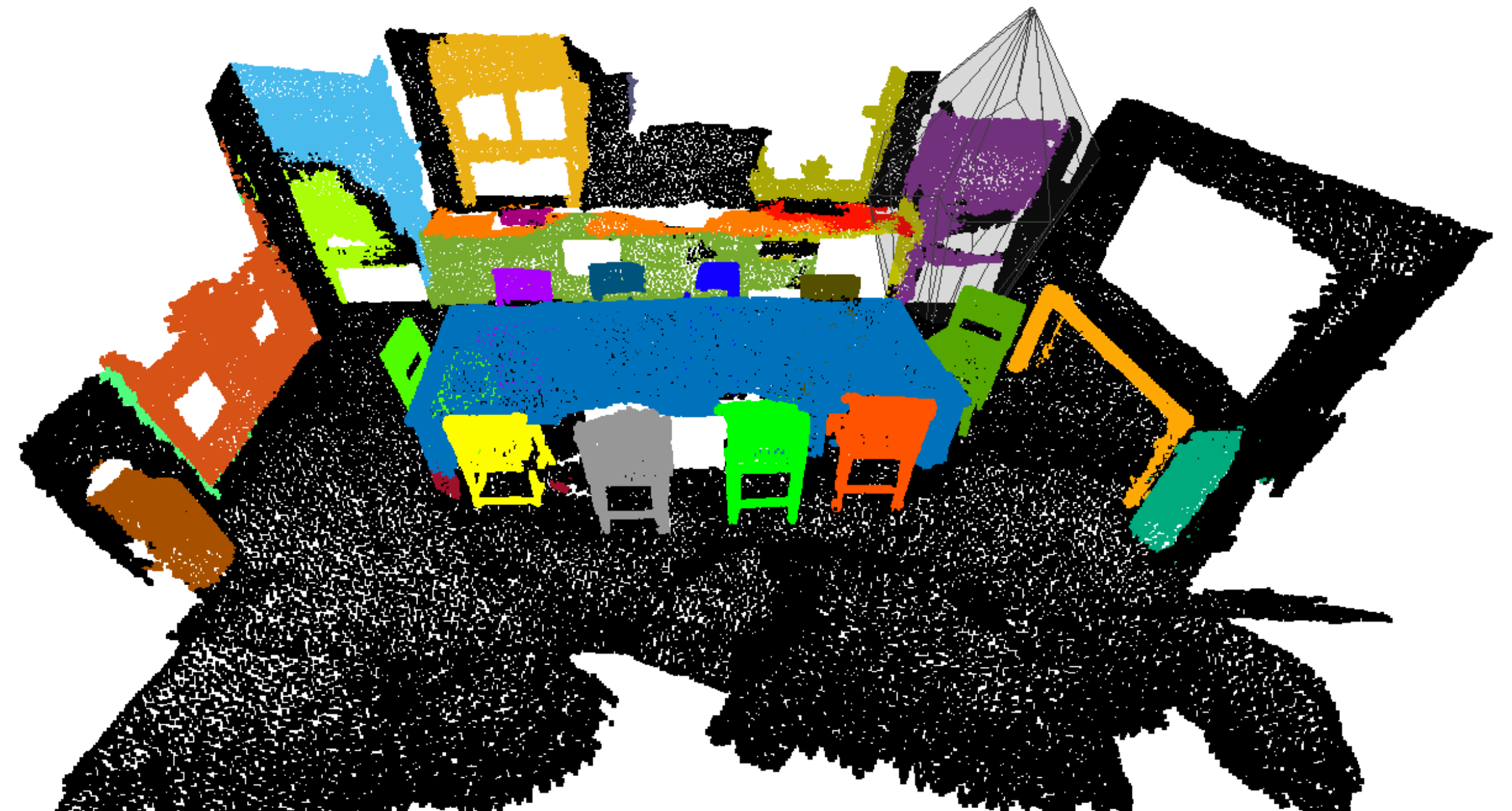


Results — Learned convex hull of high quality

Convex hulls in 2D



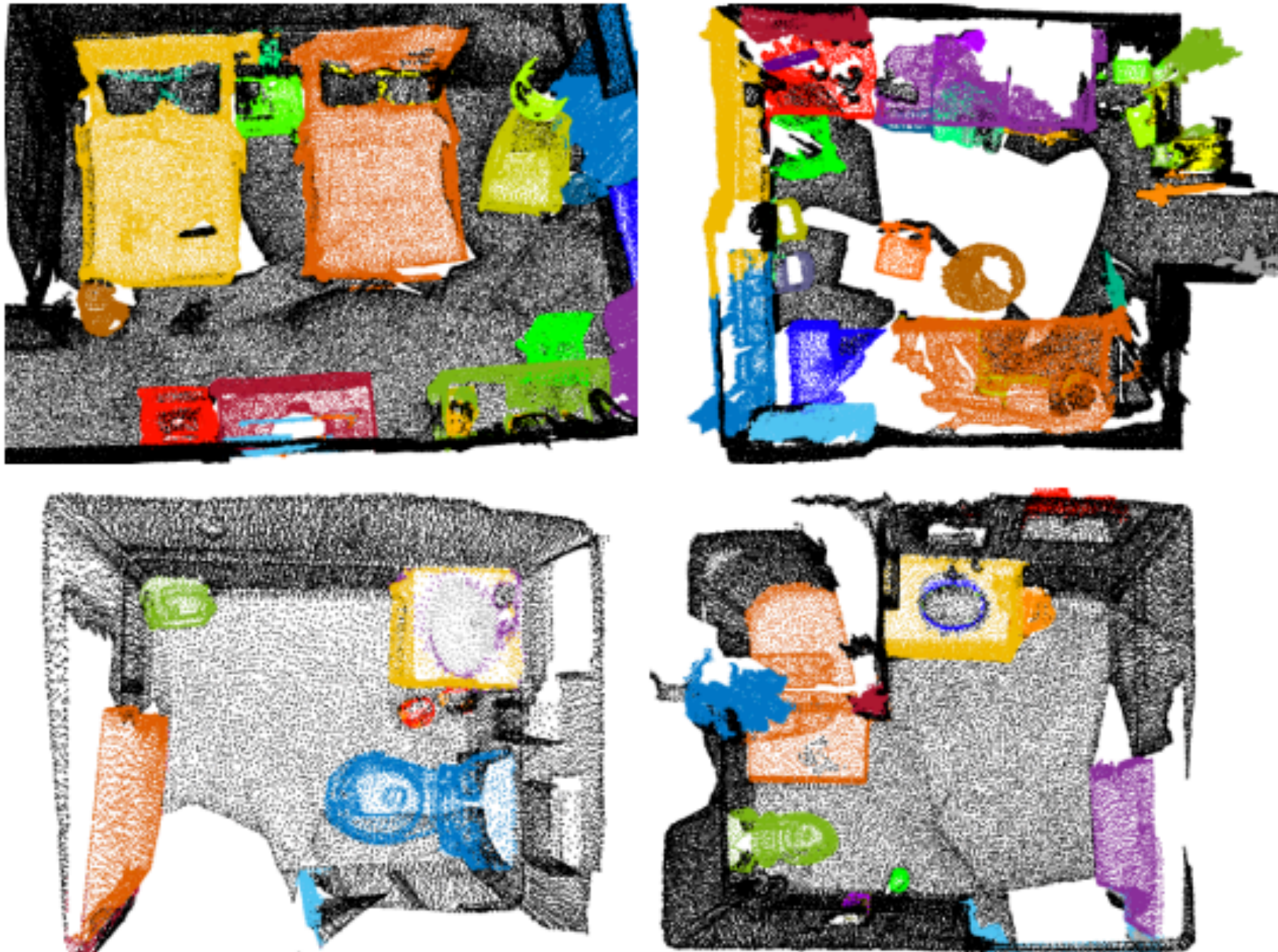
Convex hulls in 3D



Results — Instance segmentation in ScanNet.

- Our method is the **best among purely top-down** methods
- Top-down methods in general lags behind the bottom-up/Mix strategy

Qualitative



Quantitative

Methods		Validation			Test		
		mAP	AP ₅₀	AP ₂₅	mAP	AP ₅₀	AP ₂₅
Bottom-up	PointGroup [22]	34.8	56.7	71.3	40.7	63.6	77.8
	SSTNet [26]	49.4	64.3	74.0	50.6	69.8	78.9
	HAIS [2]	43.5	64.4	75.6	45.7	69.9	80.3
Mix	Dyco3D [18]	35.4	57.6	72.9	39.5	64.1	76.1
	SoftGroup [44]	—	67.6	78.9	50.4	76.1	86.5
Top-down	3D-SIS [20]	—	18.7	35.7	16.1	38.2	55.8
	GSPN [53]	19.3	37.8	53.4	—	30.6	—
	3D-Bonet [52]	—	—	—	25.3	48.8	68.7
	Ours	36.0	55.5	71.1	35.3	55.5	71.8

Summary



Explainable models have lots of potential. When carefully designed, these models can also have good performance!

wsunid.github.io